

# Karthik Bhattaram

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## EDUCATION

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### University of California, Santa Barbara

*Master of Science in Computer Science*

Santa Barbara

*Jan. 2026 to Mar. 2027*

- Area of Focus: Machine Learning, Network Security, Systems

### University of California, Santa Barbara

*Bachelor of Science in Computer Science*

Santa Barbara

*Sept. 2022 to Dec. 2025*

- Courses: Machine Learning for Networking, Computer Security, Data Structures and Algorithms
- Honors: Regents' Scholar, College of Engineering Honors, USACO Silver Division

## EXPERIENCE

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### Software Engineer Intern

*Arista Networks*

June 2025 – Sept. 2025

*Santa Clara, California*

- Built C++ utilities to parse and interpret hardware registers in Broadcom's switching ASICs
- Created a custom data structure to detect deadlocked Ethernet interfaces and mitigate **PFC pause storms**

### Machine Learning Researcher

*University of California, Santa Barbara*

Sept. 2024 – June 2025

*Santa Barbara*

- Developed pipelines to automate common network measurement efforts
- Aggregated FCC datasets into a SQL database to enable agentic **retrieval-augmented generation** (RAG)

### Software Engineer Intern

*Cadence Design Systems / OpenEye Scientific*

July 2024 – September 2024

*Santa Fe, New Mexico*

- Wrote **AWS**-based package to benchmark database compression algorithms
- Updated **CUDA** C++ software using new language features to match competing solutions

### Hardware Architecture Intern

*Cadence Design Systems*

June 2023 – Sept. 2023

*San Jose, California*

- Developed an automation tool to produce memory wrappers of various dimensions and architectures
- Collected **System Verilog** files to be used as templates for this tool and tested memories it produced

### Software Engineer Intern

*Axiado Corporation*

June 2021 – Aug. 2022

*San Jose, California*

- Developed a package to create **network packet flows** based on IPv6, ARP, ICMP, etc. in simulation
- Created and executed a test plan of a hardware firewall's parser module and discovered multiple bugs

## PROJECTS

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### College Course Database | *Java, SQL*

March 2025 – May 2025

- Designed a database schema using Entity-Relationship diagrams
- Wrote SQL commands to create the tables and enforce data integrity
- Developed a JDBC backend to handle database transactions
- Created distinct command line interfaces for the registrar and students

## TECHNICAL SKILLS

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**Certificates:** Enterprise Security Professional, Data Science and AI

**Networking Technologies:** Scapy, Wireshark, TCP/IP, Ethernet

**Languages:** Python, C/C++, Java, JavaScript, Shell, Verilog

**Developer Tools:** Git, Visual Studio Code, Vim, Emacs

**Libraries:** CUDA, Pandas, NumPy, PyTest, Black, Matplotlib